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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Powered Smart Agriculture Monitoring and Crop Disease Prediction Platform	
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Abstract

AI-Powered Smart Agriculture Monitoring and Crop Disease Prediction Platform

Agriculture plays a vital role in ensuring food security and economic growth, yet farmers continue to face challenges such as delayed crop disease detection, unpredictable weather conditions, inefficient irrigation practices, and limited access to real-time agricultural insights. Traditional methods of crop monitoring often rely on manual inspection, which is time-consuming and may result in reduced crop yield and increased production costs. To address these challenges, this project proposes an **AI-Powered Smart Agriculture Monitoring and Crop Disease Prediction Platform** that leverages Artificial Intelligence, Machine Learning, and cloud-native technologies to support precision farming.

The proposed platform enables farmers to upload crop leaf images for automated disease detection using deep learning models while also providing real-time monitoring of environmental conditions such as temperature, humidity, and soil moisture through integrated sensor data. Based on these inputs, the system generates intelligent recommendations for irrigation, fertilizer usage, and disease management to improve crop health and productivity.

The platform also offers intelligent recommendations for irrigation scheduling, fertilizer application, crop selection, and preventive measures based on environmental data and crop health status. Farmers can monitor crop conditions through an interactive dashboard that presents disease reports, environmental analytics, historical predictions, and personalized farming recommendations. An administrative module enables agricultural experts to manage datasets, monitor system performance, update disease information, and oversee user activities.

The platform follows modern **Software Engineering** practices and adopts **DevOps**, **DevSecOps**, and **MLOps** principles to ensure reliable development, secure deployment, and continuous model improvement. It incorporates containerization using Docker, automated CI/CD workflows with GitHub Actions, cloud deployment on AWS, performance monitoring with Prometheus and Grafana, and machine learning lifecycle management through MLflow. By integrating predictive analytics, scalable cloud infrastructure, and continuous monitoring, the proposed system provides an efficient, secure, and intelligent solution for sustainable agriculture and precision farming.

Signature of the Guide

HOD